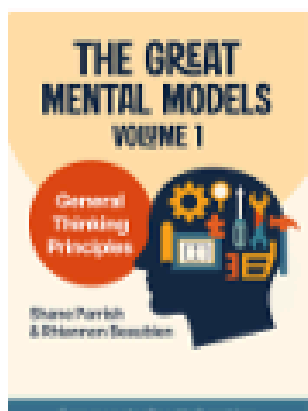




THE GREAT MENTAL MODELS, VOLUME 1

General Thinking Principles

Shane Parrish & Rhiannon Beaubien



The Big "So What"

The quality of your thinking shapes the quality of your life. In today's fast-moving world, those who think clearly hold an advantage. This book presents 9 proven thinking tools used by top leaders, scientists, and strategists to solve complex problems and navigate uncertainty. You too can use them to solve real-world problems more effectively.



What are Mental Models—And Why They Matter



Mental models are shortcuts we use to understand, decide, and act on real-world situations. Like lenses, they shape how we interpret reality and respond to it.

- Poor or outdated mental models lead to poor outcomes. Often, we fall into 3 predictable thinking traps:
 - (i) Limited perspective: we get stuck in our own viewpoint and miss the bigger picture.
 - (ii) Ego: we cling to being "right," even when facts prove us wrong.
 - (iii) Distance from consequences: we don't learn from our mistakes when we don't feel the real impact of our actions.
- To think better, we must overcome these cognitive flaws—by adopting better models. The best thinkers don't just use one framework. They draw from a network of ideas across disciplines (like physics, biology, psychology, and economics), choosing the right model for the right situation, and updating their thinking over time.

KEY QUOTES

"In life and business, the person with the fewest blind spots wins."

"The quality of your thinking depends on the models that are in your head."

"Mental models describe the way the world works. They shape how we think, how we understand, and how we form beliefs."

"Thinking better isn't about being a genius. It is about the processes we use to uncover reality and the choices we make once we do."

"When you learn to see the world as it is, and not as you want it to be, everything changes. The solution to any problem becomes more apparent when you can view it through more than one lens."



This book is the first in a multi-volume series by Shane Parrish and Rhiannon Beaubien.

- The authors distill timeless insights from various disciplines into mental models and practical tools that anyone can use to avoid common thinking traps, learn faster and make smarter choices.
- In this summary, we'll explore 9 essential models for general thinking, followed by 3 supporting concepts to help you think more clearly across all areas of life.



The 9 Fundamental Thinking Concepts



Mental Model 1: Reality > Model



*Models are simplifications, not reality.
Use them wisely, without mistaking
them for the truth.*



Every day, we use maps, models and frameworks to make sense of the world, from city layouts to business frameworks. **Models simplify reality**, allowing us to navigate complexity, make predictions and take action.



However, **all models are imperfect**. Their usefulness lies in their ability to help us understand and respond more effectively to the world around us, but they cannot capture reality perfectly.

KEY QUOTES

"I believe in the discipline of mastering the best of what other people have figured out." – Charlie Munger

"This book is about the pursuit of...wisdom, the pursuit of uncovering how things work, the pursuit of going to bed smarter than when we woke up."

*"All models are wrong; the practical question is how wrong do they have to be to not be useful."
–George Box*

- Philosopher Alfred Korzybski used the phrase “the map is not the territory” to emphasize that a model may resemble the structure of reality but never fully capture it.
 - For example, maps are designed to simplify reality, often omitting variables or distorting scale. A subway map for passengers will be useless to train operators or someone trying to navigate above-ground. A map that included everything would be too complex to use.
- Newtonian physics was used for centuries to predict how objects move, but it broke down when applied to things like light or gravity on cosmic scales. That’s where Einstein’s theory of relativity offered more accurate predictions.
- Garrett Hardin’s “Tragedy of the Commons” says that shared resources are always overused when individuals act selfishly. This was a widely accepted model in social sciences, until Elinor Ostrom later showed how communities can successfully manage shared resources, i.e. the original model didn’t apply universally.



We run into problems when we confuse models with reality, cling to outdated models, or treat abstract frameworks as universal truths. This leads us to make critical mistakes, or miss emerging risks and opportunities. If your GPS tells you to drive off a cliff, it’s up to you to check reality instead of trusting it blindly.



Use models to guide your decisions, without following them rigidly. Apply these 3 principles to use models more effectively:

- Always test models against real-world outcomes, and update your model as reality changes. For example, Frederick Taylor’s “scientific management” model was effective in factories, but it fails in today’s knowledge economy, where autonomy, learning, and purpose matter more.
- Consider who made the model and why. Every map reflects the values and assumptions of its creator, and omits things for a reason. Colonial maps, for instance, often prioritized political control over cultural accuracy.
- Watch how models shape behavior. When we follow maps or models too literally, we can create unintended outcomes—often for the worse. For example, urban planners once

KEY QUOTES

“Territories change, sometimes faster than the maps and models that describe them.”

“The description of the thing is not the thing itself. The model is not reality. The abstraction is not the abstracted.”

“The map appears to us more real than the land.”
— D.H. Lawrence

“When we mistake the map for reality, we start to think we have all the answers.”

“Maps are useful to the extent they are explanatory and predictive.”

“If reality has changed the map must change.”

“Maps are not purely objective creations. They reflect the values, standards, and limitations of their creators.”

- Talk to a “Lifer” or an expert inside the domain. Find out how they think, not just what to do.
 - However, beware of vested interests when seeking advice, especially from people whose income or success depends on the decision you make. Always ask: What does this person stand to gain?
- Use general thinking models (like inversion or second-order thinking) to reason through unfamiliar territory and make better choices.
- Take Queen Elizabeth I, for example. When she first took the throne, she recognized her inexperience. So, she built a trusted council of advisors, used their inputs to guide her decisions, and laid the foundation for a long and stable reign.



Mental Model 3: Think From the Ground Up



*Break things down to basics.
Then rebuild ground up from
First Principles.*



Don't just copy what works or tweak what already exists. Instead, break things down into First Principles (i.e. foundational truths in a specific context) then build up from there.

- In physics, that might be thermodynamic laws. In food, it might be asking "What makes something 'meat'—is it the fact that it comes from an animal—or the taste, texture, and nutritional makeup we associate with it?"



Use 2 methods to uncover first principles:

- Challenge assumptions through Socratic questioning or structured inquiry. Start by clarifying your position, e.g. "What do I think and why?" Then, probe deeper:
 - What assumptions am I making?
 - What's the evidence for this belief?
 - Are there alternative explanations?
 - What are the implications if I'm wrong?

KEY QUOTES

"Whenever we are getting advice, it is from a person whose set of incentives is not the same as ours."

*"Ignorance more often begets confidence than knowledge."
— Charles Darwin*

"First principles thinking is one of the best ways to reverse-engineer complicated situations and unleash creative possibility."

"Everything that is not a law of nature is just a shared belief. Money is a shared belief. So is a border. So are bitcoin. So is love."

"If we never learn to take something apart, test our assumptions about it, and reconstruct it, we end up bound by what other people tell us—trapped in the way things have always been done."

KEY QUOTES

*"Science is much more than a body of knowledge. It is a way of thinking."
— Carl Sagan*

"If you know the first principles of something, you can build the rest of your knowledge around them to produce something new."

*"As to methods, there may be a million and then some, but principles are few."
— Harrington Emerson*

"You can change the tactics if you know the principles."

"A detailed thought experiment...lets us do things in our heads we cannot do in real life, and so explore situations from more angles than we can physically examine and test for."

"Thought experiments are more than daydreaming. They require the same rigor as a traditional experiment in order to be useful."

- The Five Whys: Start with an issue and ask "Why is this happening?" Repeat the question 5 times or until you land on a root cause (expressed as a "what" or "how").
 - Example: A product fails. Why? → The battery overheated. Why? → A component failed... and so on, until you uncover a design flaw or neglected test.



In other words, don't just copy what works; understand why it works so you're no longer limited by how things have always been done.

- For decades, doctors believed the stomach was sterile, until Robin Warren and Barry Marshall challenged the assumption—and discovered that stomach ulcers were caused by bacteria. This overturned decades of misdiagnosis and ineffective treatments, transforming how ulcers are now treated.
- Most slaughterhouses used straight chutes simply because that's how it had always been done. Temple Grandin challenged this by studying animal behavior and identifying a first principle: reducing stress is key to moving cattle humanely and efficiently. This led to innovations like curved chutes that blocked visual distractions, though any design would work as long as it addressed the underlying principle of stress reduction.



Mental Model 4: Test It In Your Mind



When you can't test something in real life, simulate it mentally.



Some ideas or decisions are too costly, risky, or complex to test in the real world. That's where thought experiments come in. By simulating possibilities in our minds, we can tweak the variables and examine possible outcomes to stretch our thinking and make smarter choices.

- A good thought experiment is structured and intentional:
 - (i) Ask a clear, focused question.
 - (ii) Do background research to understand the context.

- (iii) Form a hypothesis: What do you expect to happen?
- (iv) Run the scenario mentally, tweaking key variables to explore different outcomes.
- (v) Analyze the outcomes, and refine your conclusion or assumptions before taking action.



Thought experiments are especially useful in 3 situations:

- Simulating physical scenarios that can't be recreated in a lab. When developing his theory of relativity, Albert Einstein imagined being in a sealed elevator in space. Would he feel a force pulling him down (gravity) or pushing him up (acceleration)? Since both felt identical from inside the elevator, he concluded the effects were equivalent—an insight that became the foundation of general relativity.
- Re-imagine history by exploring "what might have happened" if certain factors were changed. For instance, the assassination of Archduke Franz Ferdinand was considered a key event that triggered World War I. Without it, would the war still have happened? By researching and imagining how other variables might have played out differently, we gain a better understanding of the relative impact of each factor.
 - However, remember that in complex systems like history, small changes can lead to drastically different outcomes.
- Testing whether our gut-intuition is robust. Philosopher John Rawls proposed a thought experiment to design a fair society: Imagine you don't know your future position in that society—whether you'd be rich or poor, powerful or powerless. Would you still choose the same rules? This "veil of ignorance" strips away self-interest and reveals what fairness really looks like.

KEY QUOTES

"One of the real powers of the thought experiment is that there is no limit to the number of times you can change a variable to see if it influences the outcome."

"A small change in the past could have a massive, unpredictable effect on the future."

"Thought experiments tell you about the limits of what you know and the limits of what you should attempt."



Mental Model 5: Consider Ripple Effects



Look beyond immediate steps. Consider the ripple effects and what those ripple effects might cause in turn.



People tend to focus on the obvious, immediate outcome of an action (first-order effect). But every action has **additional consequences (second-order and beyond)** that are more impactful in the long run.

- First-order thinking is linear and short-term: "If I eat this candy, I'll enjoy it."
- Second-order thinking looks long-term: "If I keep eating candy every day, what will my health look like in 10 years?"
 - It goes beyond surface-level thinking to examine (i) short-term wins vs long-term costs, (ii) unintended side effects and (iii) chain reactions that affect other systems, trigger new problems or tradeoffs.
 - This is crucial when dealing with complex systems like markets, ecosystems, organizations, or relationships, where everything is interconnected.



Second-order thinking is **especially useful for 2 things**:

- Weighing short-term vs long-term consequences, to avoid creating long-term pains just for short-term gratification. For example, antibiotics have been added to livestock feed for years to prevent disease and to accelerate growth. This makes meat cheaper and production more efficient (first-order benefits). But over time, it also led to the rise of antibiotic-resistant bacteria—a second-order effect that now threatens human health.
- Make better arguments by thinking ahead. Rather than arguing only for women's rights on moral grounds, philosopher Mary Wollstonecraft addressed second-order benefits, arguing that educating women would lead to stronger families and a better-functioning society.

KEY QUOTES

"You can never merely do one thing."
—Garrett Hardin

"We must ask ourselves the critical question: And then what?"

"Second-order thinking involves asking ourselves if what we are doing now is going to get us the results we want."

"We operate in a world of multiple, overlapping connections, like a web, with many significant, yet obscure and unpredictable, relationships."

"Maximizing benefits is something that happens over time."

"A little time spent thinking ahead can save us massive amounts of time later."

"We don't make decisions in a vacuum and we can't get something for nothing."

- Marie Van Brittan Brown felt unsafe at home. Rather than adding more locks, she asked: What would make me feel safe before opening the door? Her answer led to the invention of the home CCTV system, laying the groundwork for modern home security.

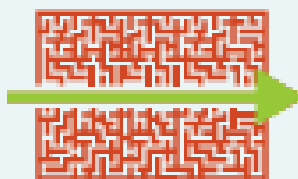


Inversion is especially powerful when you feel stuck or lost.

- Define what you want to avoid, then eliminate the conditions that would cause it—like how John Bogle created the index fund by seeking to avoid investment losses.
- If success seems impossible, ask “What must be true for it to happen?” or start with an assumption. When Sherlock Holmes needed to find a hidden object, he assumed “If she truly values it, she’ll try to protect it during an emergency.” He then staged a fake fire—and observed where the person instinctively went, revealing the object’s location without a direct search.



Mental Model 8: Start with the Simplest Explanation



When in doubt, start with the simplest explanation (with fewest assumptions) that fits the facts.



Occam’s Razor reminds us to start with the simplest explanation—one that makes the fewest assumptions. More assumptions means higher risks of being wrong.

- Simple models are easier to understand, test, adjust, and they’re more robust in fast-moving or uncertain environments. That’s because every layer of complexity or moving part introduces potential points of failure.
- Cut away what doesn’t matter to help you see what does:
 - In *medicine*, doctors use this model to start with the most likely diagnosis. If a patient shows flu-like symptoms, the simplest explanation is the flu—not Ebola.
 - When *IBM* was in crisis during the early 1990s, CEO Louis Gerstner rejected calls for a grand vision. Instead, he focused on simple, tough-minded execution: serving existing customers and building profitable businesses. This saved the company without any complicated overhauls.

KEY QUOTES

“Simply invert, always invert, when you are stuck. If you take the results of your inversion seriously, you might make a great deal of progress on solving your problems.”

“The simplest explanation is almost certainly the correct one.”

“We should prefer the simplest explanation with the fewest moving parts. They are easier to falsify, easier to understand, and generally more likely to be correct.”

“When you hear hoofbeats, think horses, not zebras.”
— A medical saying

“The last thing IBM needs right now is a vision.”
—Louis Gerstner

✔ Obviously, some truths are more complex. The key is to start simple, then add complexity when needed and justified by evidence.



Mental Model 9: Assume Positive Intent



Don't assume ill intent when ignorance or incompetence will do.

🧠 Hanlon's Razor reminds us not to jump to conclusion about people's intentions, as it can lead to conflict and misjudgment.

- In an experiment, people were told that "Linda is outspoken and deeply concerned with social justice." Most people incorrectly judged it more likely that she was both a feminist and a bank teller, even though logically, she had to be more likely to be a bank teller alone. This shows how we can make illogical assumptions about intention and identity.
- When things go wrong, don't assume it's due to malice. More often, it's due to carelessness, ignorance, or flawed systems. If someone spills a drink on you at a party, it's more likely to be an accident than an intentional attempt to ruin your night.

🧠 Real malice does exist. But in most situations, assuming positive intent helps us to stay calm, prevent paranoia, avoid unnecessary conflict, and make better decisions—especially when misjudging intent can cause more harm than the mistake itself. For example:

- In 1962, Soviet officer Vasili Arkhipov was aboard a submarine during the Cuban Missile Crisis. When the Americans dropped depth charges, the crew assumed they were under attack. The captain wanted to retaliate with a nuclear torpedo. But Arkhipov stayed calm and insisted on surfacing and contacting Moscow—a decision that prevented a nuclear war.
- In 408 AD, Roman emperor Honorius misjudged his loyal general Stilicho and had him executed, believing he was conspiring against him. This fatal error left the Western Roman Empire without its most capable defender—and it soon collapsed.

KEY QUOTES

*"Anybody can make the simple complicated. Creativity is making the complicated simple."
—Charles Mingus*

"We should not attribute to malice that which is more easily explained by stupidity."

"If we present the evidence in a certain light, the brain malfunctions. It doesn't weigh out the variables in a rational way."

"There are fewer true villains than you might suppose...all humans make mistakes and fall into traps of laziness, bad thinking, and bad incentives."

"Most people doing wrong are not bad people trying to be malicious....Always assuming malice puts you at the center of everyone else's world. This is an incredibly self-centered approach to life."

"When we assume someone is out to get us...[it] tends to reduce our vision to dealing with the perceived threat instead of examining the bigger picture."



3 Supporting Ideas for Better Reasoning



Falsifiability: To be scientific, a theory must be testable, i.e. it can be disproved by statements like "If X happens, theory Y is false." Otherwise, it's just a belief or an opinion.

- For instance, Freudian psychology is hard to prove or disprove, since it can explain any outcome after the fact.
- In contrast, evolution by natural selection makes claims that are specific and testable. For example, if traits harmful to survival routinely spread through populations, it would falsify the theory of natural selection. Repeated testing continues to support the theory so far.



Necessary $\neq$ Enough: Don't confuse necessity with sufficiency. Necessary conditions are what you must have to succeed, but they don't guarantee success. For example, skill and effort might be necessary for success, but they're rarely sufficient. Other factors (like timing, competition, or luck) also play a role.



Causation vs. Correlation: Just because two things occur together doesn't mean one causes the other.

- Correlation does not imply causation. For example, a study might show that children of heavy drinkers perform worse in school. But does drinking cause poor grades? Does stress from a child's struggles cause parents to drink more? Or are other factors responsible?
- Understand the strength and direction of correlation to avoid overgeneralizing or misapplying data:
 - Perfect correlation (1): This means they measure the same thing in different units (e.g. Celsius and Fahrenheit).
 - Zero (0) correlation, e.g. bottled water consumption and suicide rates have no relation.
 - Weak/moderate correlation (e.g. 0.3): Height and weight correlate somewhat, but other factors matter too.

KEY QUOTES

"Extraordinary claims require extraordinary proof."
—Carl Sagan

"The set of conditions necessary to become successful is a part of the set that is sufficient to become successful. But the sufficient set itself is far larger than the necessary set."

"We notice two things happening at the same time (correlation) and mistakenly conclude that one causes the other (causation)."